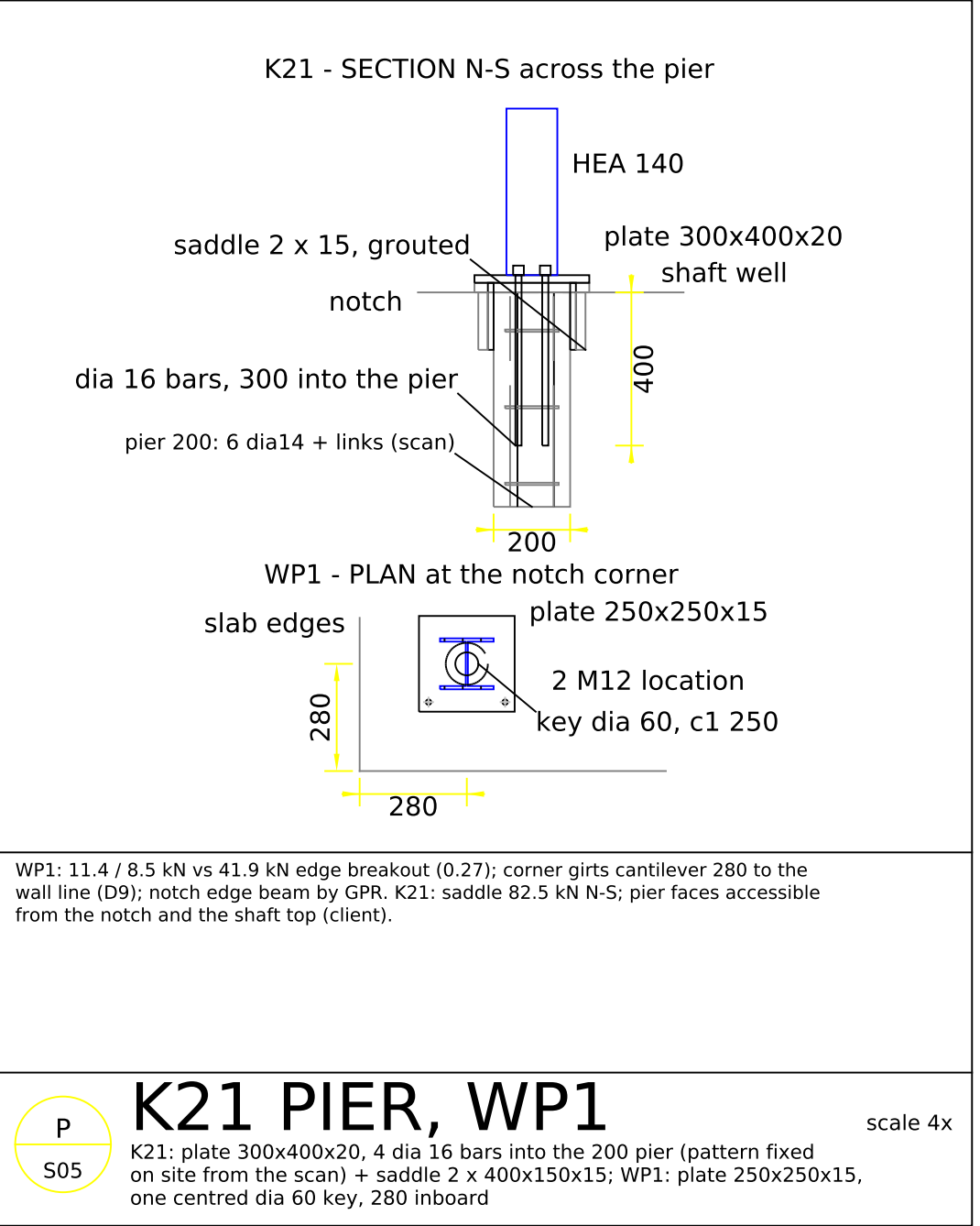
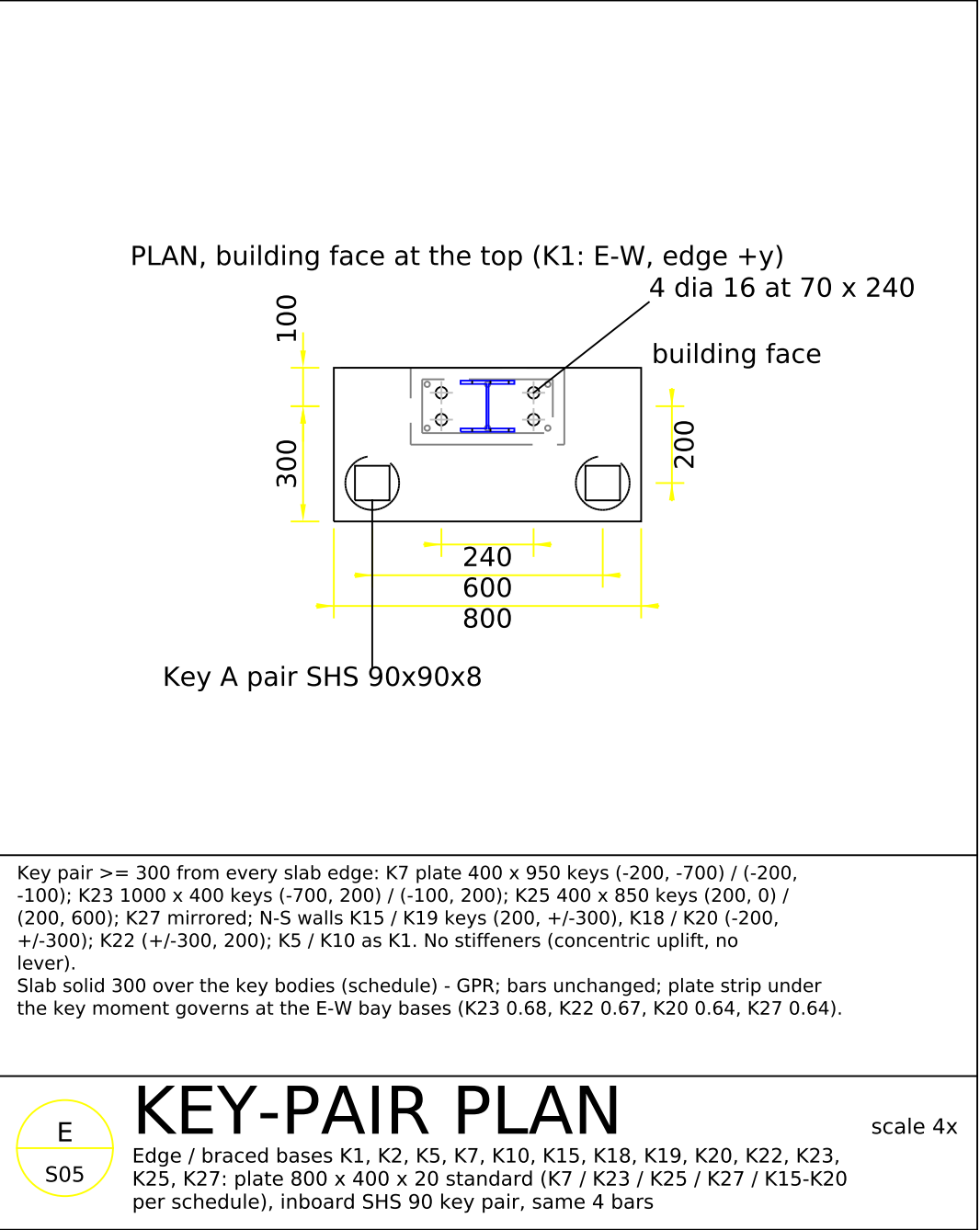
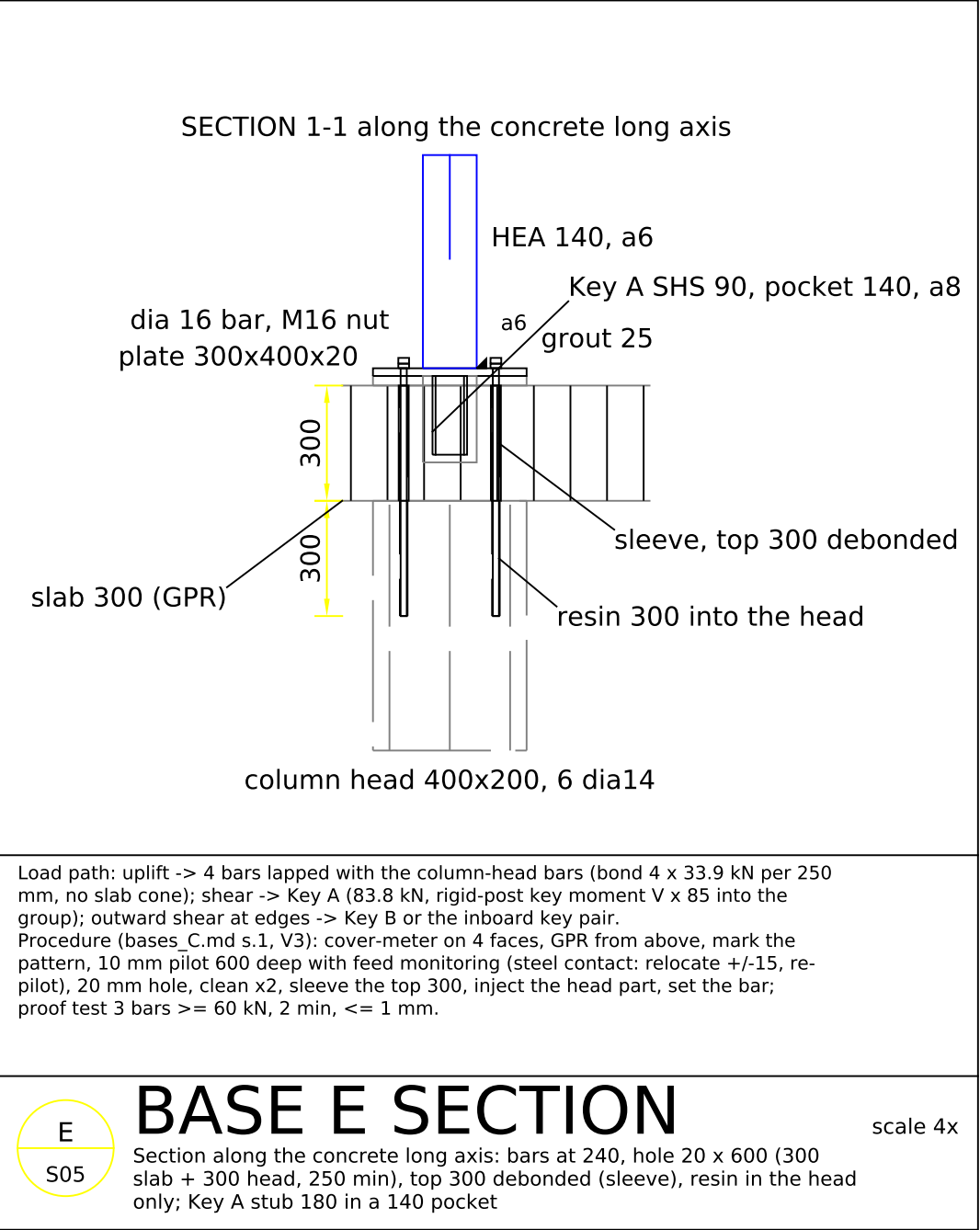
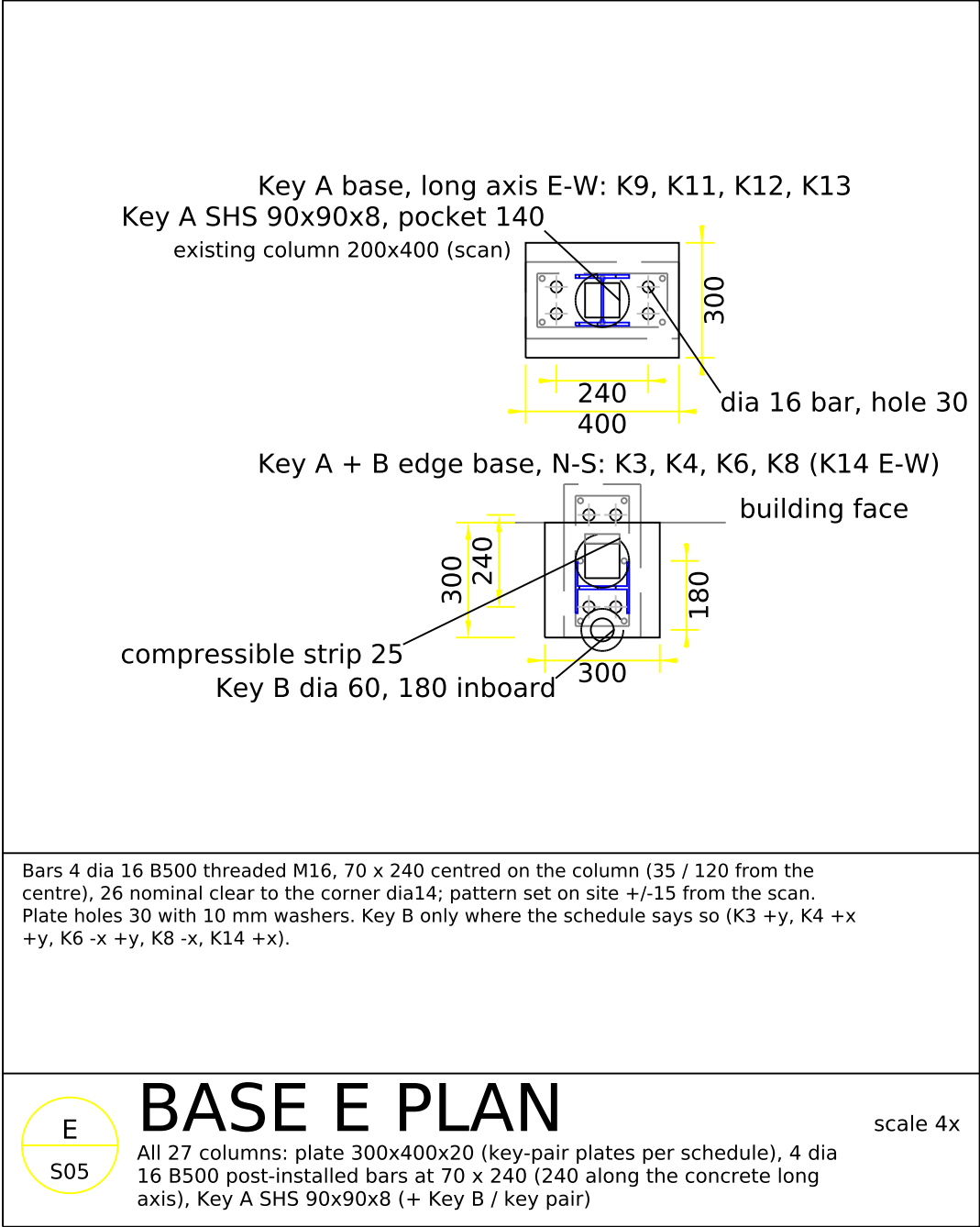




BASE SCHEDULE (bases_C.md Rev 8 s.3): x E-W, y N-S; all type E, P at K21; worst K7 0.83

Mark	K	Ty.	Axis	Bars (mm from centre)	Keys (mm from centre)	Plate (extents from centre)	Util.	Governing check
C1	K1	E	E-W	4 d16 (+/-120, +/-35)	A pair (-300,-200)/(300,-200)	800x400x20 (x -400..400, y -300..100)	0.43	plate strip at key moment
C2	K2	E	E-W	4 d16 (+/-120, +/-35)	A pair (-300,-200)/(300,-200)	800x400x20 (x -400..400, y -300..100)	0.43	plate strip at key moment
C3	K3	E	N-S	4 d16 (+/-35, +/-120)	A centre + B +y	300x400x20	0.37	Key B outward +y
C4	K4	E	N-S	4 d16 (+/-35, +/-120)	A centre + B +x,+y	300x400x20	0.36	Key B outward +x
C5	K5	E	E-W	4 d16 (+/-120, +/-35)	A pair (-300,-200)/(300,-200)	800x400x20 (x -400..400, y -300..100)	0.65	plate strip at key moment
C6	K6	E	N-S	4 d16 (+/-35, +/-120)	A centre + B -x,+y	300x400x20	0.39	Key B outward -x
C7	K7	E	N-S	4 d16 (+/-35, +/-120)	A pair (-200,-700)/(-200,-100)	400x950x20 (x -300..100, y -800..150)	0.88	E key pair bearing
C8	K8	E	N-S	4 d16 (+/-35, +/-120)	A centre + B -x	300x400x20	0.37	Key B outward -x
C9	K9	E	E-W	4 d16 (+/-120, +/-35)	A centre	300x400x20	0.28	E rebar bond in the column head, embed
C10	K10	E	E-W	4 d16 (+/-120, +/-35)	A pair (-300,-200)/(300,-200)	800x400x20 (x -400..400, y -300..100)	0.19	E rebar bond in the column head, embed
C11	K11	E	E-W	4 d16 (+/-120, +/-35)	A centre	300x400x20	0.19	E rebar bond in the column head, embed
C12	K12	E	E-W	4 d16 (+/-120, +/-35)	A centre	300x400x20	0.33	E rebar bond in the column head, embed
C13	K13	E	E-W	4 d16 (+/-120, +/-35)	A centre	300x400x20	0.28	E rebar bond in the column head, embed
C14	K14	E	E-W	4 d16 (+/-120, +/-35)	A centre + B +x	300x400x20	0.57	Key A parallel to edge +x
C15	K15	E	N-S	4 d16 (+/-35, +/-120)	A pair (200,-300)/(200,300)	400x800x20 (x -100..300, y -400..400)	0.50	plate strip at key moment
C16	K16	E	N-S	4 d16 (+/-35, +/-120)	A centre	300x400x20	0.04	E rebar bond in the column head, embed
C17	K17	E	N-S	4 d16 (+/-35, +/-120)	A centre	300x400x20	0.04	E rebar bond in the column head, embed
C18	K18	E	N-S	4 d16 (+/-35, +/-120)	A pair (-200,-300)/(-200,300)	400x800x20 (x -300..100, y -400..400)	0.37	plate strip at key moment
C19	K19	E	N-S	4 d16 (+/-35, +/-120)	A pair (200,-300)/(200,300)	400x800x20 (x -100..300, y -400..400)	0.50	plate strip at key moment
C20	K20	E	N-S	4 d16 (+/-35, +/-120)	A pair (-200,-300)/(-200,300)	400x800x20 (x -300..100, y -400..400)	0.66	plate strip at key moment
C21	K21	P	E-W	4 d16 (+/-120, +/-35)	saddle	300x400x20 + saddle	0.21	saddle plates
C22	K22	E	E-W	4 d16 (+/-120, +/-35)	A pair (-300,200)/(300,200)	800x400x20 (x -400..400, y -100..300)	0.71	plate strip at key moment
C23	K23	E	E-W	4 d16 (+/-120, +/-35)	A pair (-700,200)/(-100,200)	1000x400x20 (x -800..200, y -100..300)	0.72	plate strip at key moment
C24	K24	E	N-S	4 d16 (+/-35, +/-120)	A centre	300x400x20	0.23	Key A towards edge -y
C25	K25	E	N-S	4 d16 (+/-35, +/-120)	A pair (200,0)/(200,600)	400x850x20 (x -100..300, y -150..700)	0.38	E key pair bearing
C26	K26	E	N-S	4 d16 (+/-35, +/-120)	A centre	300x400x20	0.38	Key A SHS bearing
C27	K27	E	N-S	4 d16 (+/-35, +/-120)	A pair (-200,0)/(-200,600)	400x850x20 (x -300..100, y -150..700)	0.66	plate strip at key moment
WP1	-	post	-	2 M12	one dia 60 centred	250x250x15	0.27	key edge breakout

LEGEND

- HEA 140 column
- Base plate / keys
- Existing slab / column head
- Grout / pockets

SLAB AND COLUMN-HEAD VERIFICATION (bases_C.md Rev 8 s.5-6)

No solid-zone criterion for the anchorage: the bars depend on the column head only (6 dia14, dia6 links, C25) - confirmed by the face cover-meter scan of every head and 3-6 cores for the grade. The slab must be solid (no blocks) and 300 mm thick under each base plate and around the key pockets (Key A +/- 400; key pairs: the key breakout bodies, +/- 700 along the wall for the standard pair) - GPR from above at all 27 heads, also for the slab top bars at the pockets.
A head with fewer than 4 sound corner bars or links > 200: embedment 300-350 (bond alone 4 x 33.9 kN per 250) and the lap re-checked with the scanned bars before drilling; a head that cannot be drilled: appendix B1 detail (4 M20 resin anchors h_ef 250 in the slab, GPR-verified 800 x 800 solid zone) at that head - client option for the 7 interior columns K9, K12, K13, K16, K17, K24, K26.
Before installation: scan sequence (faces from below, GPR from above, pilot drill); pockets 140 / 110 placed to cut at most one slab top bar; proof tests of 3 production bars (interior, edge, corner) to >= 60 kN held 2 min, displacement <= 1 mm; EAD 330087 injection system, certified installer; nuts snug + 1/4 turn after the grout has cured; scan sheet with the final pattern filed per head.
No through-bolts, no ceiling access, no under-slab plates (Rev 8). Grout bed 25 +/- 5 mm; columns cut to the surveyed plate-top level (S00 s.9).